Analyzing Ocean Waves Frequency Spectrum

Using Image Processing Techniques

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Project Overview

This project focuses on analyzing Synthetic Aperture Radar (SAR) images captured by the Sentinel-1 satellite in wave mode to investigate the wavelengths and directions of ocean waves, specifically in relation to the geophysical phenomenon of wind streaks. The SAR images used in this analysis are captured in 25 km by 25 km vignettes with a spatial resolution of approximately 5 m x 5 m.

Data Preprocessing

The first step in the analysis, if necessary, will involve preprocessing the image. The image may be sharpened to accentuate the edges of the waves, making them more distinguishable for further analysis. If needed, histogram equalization may also be applied to improve the contrast of the image. If deemed necessary filters may be applied to reduce noise.

Frequency and Direction Calculation

After preprocessing, the analysis will involve applying a Fourier Transform to the images. This step will decompose the image into its spatial frequency components, allowing for the identification of dominant frequencies and their corresponding directions. To enhance the clarity of Fourier peaks, the image will be divided into smaller sections using windows larger than the expected wavelength. This segmentation will help make the peaks clearer. Experimentation with window sizes and overlap will ensure optimal segmentation for the best results. This approach enables the calculation of average wave frequency and direction within each section, accounting for the non-uniform nature of wave patterns across the image. Both fast and slow wave frequencies will be identified, along with their respective directions.

Video Processing for Wave Speed (If Time Permits)

If time permits, the analysis will be extended to a video of ocean waves. The goal will be to determine wave speed by comparing the position of the waves in sequential video frames. This would involve analyzing phase shifts and tracking wave features over time. Using techniques learned in class, the distance traveled by a specific wave over a set time interval can be calculated to estimate its speed.

GUI Development

To streamline and facilitate the analysis, a graphical user interface (GUI) will be developed. The GUI will allow users to input SAR images, perform Fourier analysis, and visualize the results, including wave frequencies, directions, and, if applicable, wave speeds from video analysis.

Expected Outcomes

The goal of this project is to compute the dominant wavelengths and their directional orientations within each SAR image of ocean waves influenced by wind streaks.